

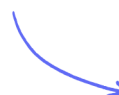
Structured Questions

Scalars & Vectors

Scalars & Vectors / Resolving Vectors

Easy (5 questions)	/46
Medium (5 questions)	/46
Hard (5 questions)	/61
Total Marks	/153

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Easy Questions

1 (a) Explain the difference between vector and scalar quantities.

(2 marks)

(b) **Table 1** contains a number of scalar and vector quantities.

Place one tick (✓) in each row to show whether each quantity is a scalar or a

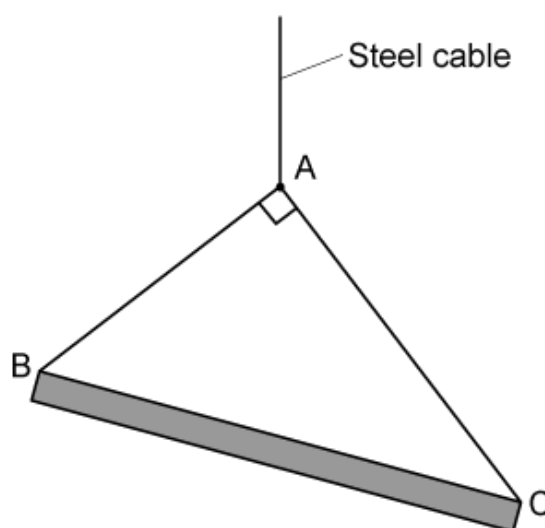
Table 1

Quantity	Scalar	Vector
Speed		
Velocity		
Mass		
Weight		
Distance		
Displacement		

(3 marks)

- (c) **Figure 1** shows a uniform beam supported by two light cables, **AB** and **AC**, which are attached to a single steel cable from a crane. The beam is stationary and in equilibrium.

Figure 1



State two reasons the beam can be described as 'in equilibrium'.

(2 marks)

- (d) The tension in the cable **AB** is 3 N and the tension in the cable **AC** is 4 N.

Calculate the resultant force required in the beam **BC** to keep the system in equilibrium.

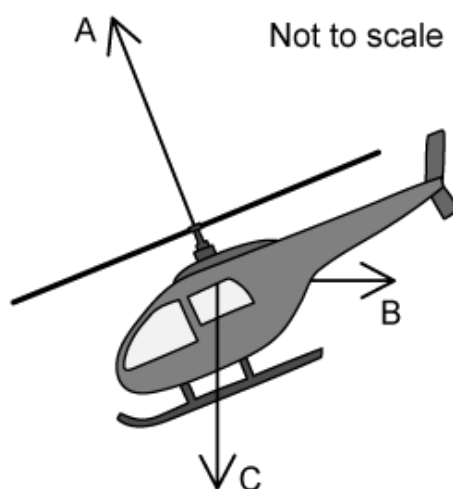
(2 marks)

2 (a) State **two** vector quantities and **two** scalar quantities.

(4 marks)

(b) The helicopter, shown in **Figure 1**, is moving horizontally through still air. The lift force from the helicopter's blades is labelled **A**.

Figure 1



Name the two forces **B** and **C** that also act on the helicopter.

(2 marks)

(c) The helicopter is moving at a constant velocity.

Draw a force vector triangle to show this arrangement.

(2 marks)

(d) The force **B** has a value of 25 kN and the force **C** has a value of 40 kN.

Calculate the value of force **A** needed to keep the helicopter moving at a constant velocity.

(2 marks)

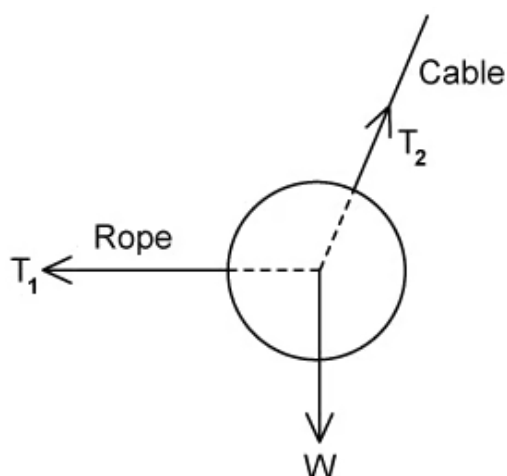
3 (a) State what is meant by a vector quantity and give one example.

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(2 marks)

(b) **Figure 1** shows a ball is suspended from a cable. The ball is pulled into the position shown by a rope that is kept horizontal.

Figure 1



In the position shown, the ball is in equilibrium.

State the force and component which balance:

- The force of the rope on the ball.
- The weight of the ball.

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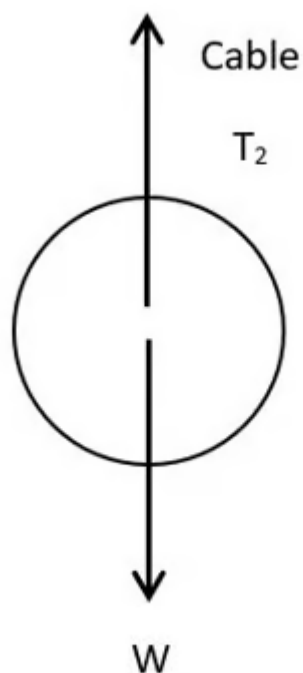
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(2 marks)

- (c) When in equilibrium, the tension T_1 is equal to 10.1 N and the tension T_2 is equal to 15.7 N.

Calculate the weight of the ball.

(2 marks)

- (d) The ball is then detached from the rope, so it is hanging only by the cable.



State the new tension in the cable.

(1 mark)

- 4 (a)** Complete **Table 1** by stating whether the quantity is a vector or a scalar and by giving the full name of its unit.

Table 1

Quantity	Vector or Scalar	Unit
Force		
Displacement		
Kinetic Energy		
Power		

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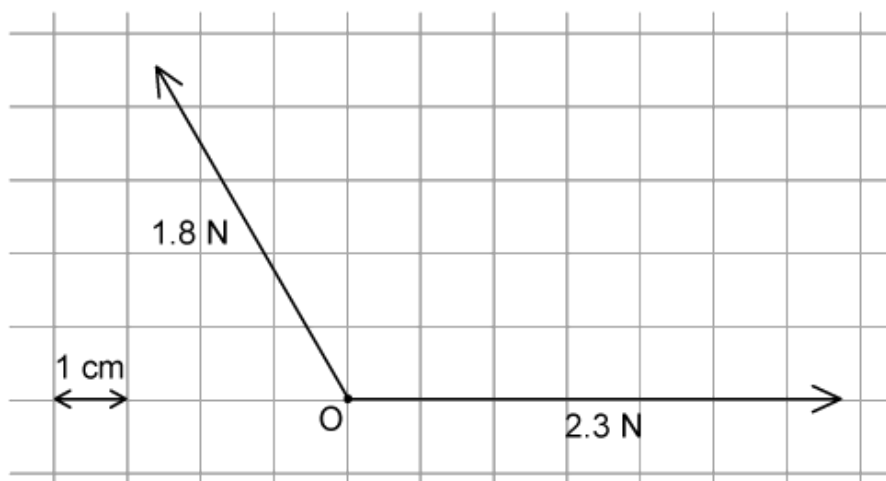
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(4 marks)

- (b) Figure 1** shows two forces acting on an object at **O**. The forces have been drawn to scale.

Figure 1



State the scale used in **Figure 1**.

(1 mark)

- (c) Complete the scale drawing, in **Figure 1**, to determine the magnitude of the resultant force.

(3 marks)

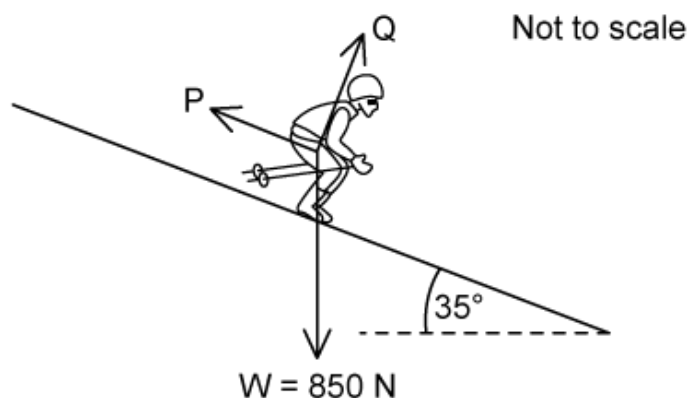
- (d) The 1.8 N force then begins to act vertically upwards, so the two forces are perpendicular to each other.

Calculate the angle at which the new resultant force acts.

(3 marks)

- 5 (a) A skier travels at constant speed down a slope of 35° , as shown in **Figure1**.

Figure1



The force labelled **P** is parallel to the slope. The force labelled **Q** is perpendicular to the slope. Assume that there is no friction between the skis and the snow.

Identify the forces labelled **P** and **Q**.

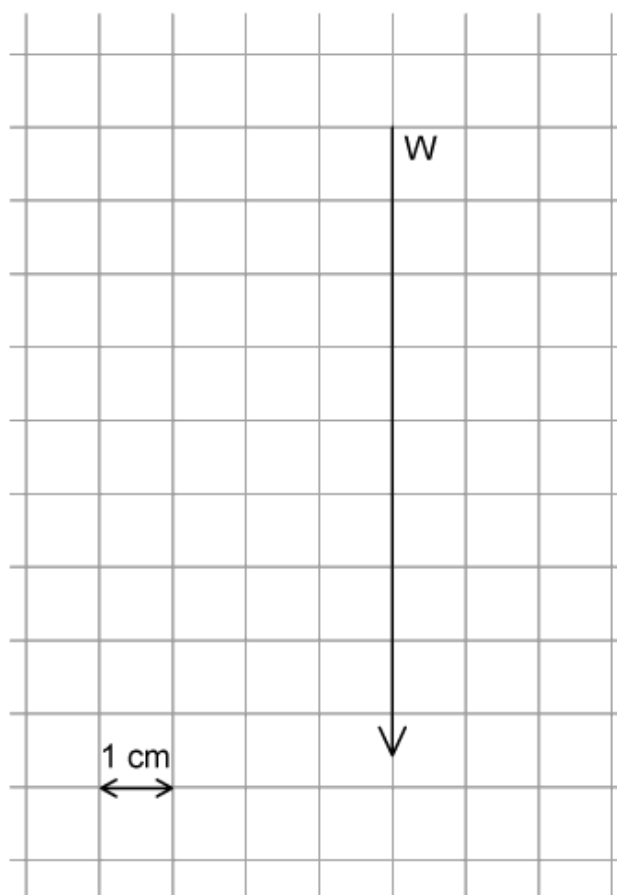
(2 marks)

- (b) State the condition necessary for the skier to be travelling at a constant velocity.

(1 mark)

- (c) **Figure 2** below shows an arrow representing the weight, **W**, of the skier. The arrow has been drawn to scale.

Figure 2



scale 1 cm: 100 N

By drawing the forces **P** and **Q** onto **Figure 2**, complete the scale diagram and determine the magnitude of the force **P**.

(4 marks)

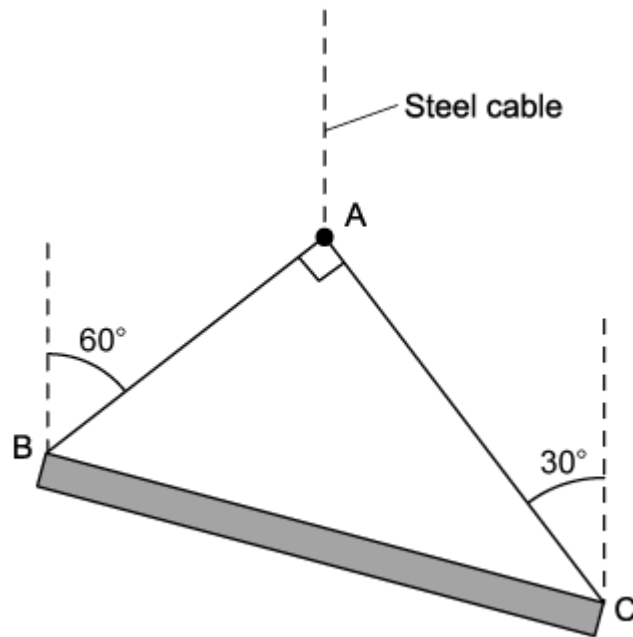
(d) Using **Figure 2**, or otherwise, determine the magnitude of force **Q**.

(2 marks)

Medium Questions

- 1 (a) **Figure 1** shows a uniform beam supported by two light cables, **AB** and **AC**, which are attached to a single steel cable from a crane. The beam is stationary and in equilibrium.

Figure 1



The tension in cable **AB** is equal to T_1 and the tension in cable **AC** is equal to T_2 . The weight of the beam is 9.5 kN.

Draw a vector diagram to show the arrangement of the forces, including the angles.

(3 marks)

- (b) Write two equations to show the vertical and horizontal components of the forces in part (a).

(2 marks)

(c) Calculate the tension T_1 in cable **AB**.

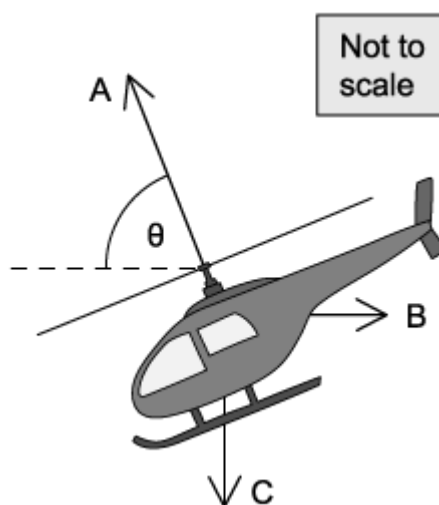
(2 marks)

(d) Calculate the tension T_2 in cable **AC**.

(2 marks)

- 2 (a) The helicopter shown in **Figure 1** is moving horizontally through still air. The lift force from the helicopter's blades is labelled **A**.

Figure 1



The lift force, **A**, is 12.0 kN and acts at an angle of θ to the horizontal. At any point, the lift force is always four times larger than the drag force.

Draw a vector diagram to show the arrangement of the forces.

(2 marks)

- (b) Show the angle θ is about equal to 76° .

(3 marks)

- (c) Calculate the weight of the helicopter. Give your answer to an appropriate number of significant figures.

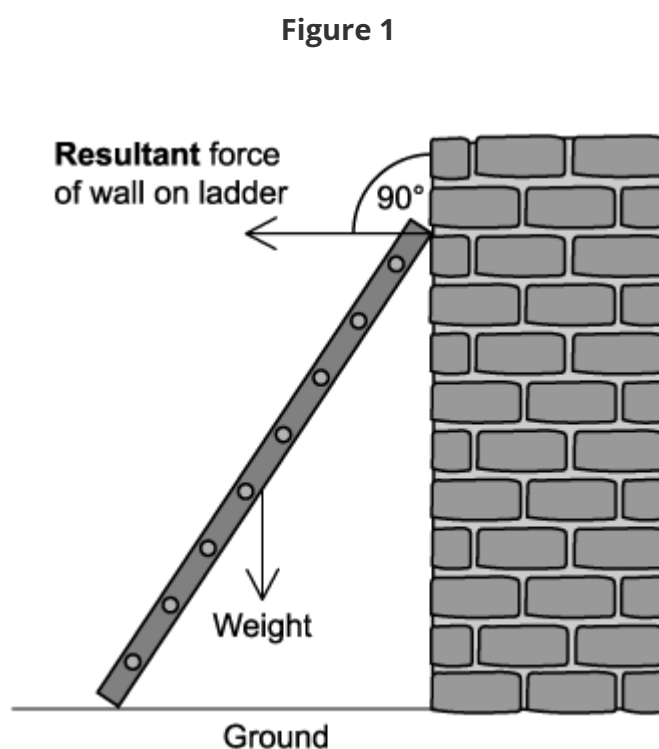
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(3 marks)

- (d)** The helicopter begins to descend at a constant speed at an angle of 15° to the horizontal.

Draw a vector triangle to represent the new forces acting on the helicopter.

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(3 marks)

- 3 (a) **Figure 1** shows two of the forces acting on a uniform ladder resting against a vertical wall.



Explain how **Figure 1** shows that the friction between the ladder and the wall is negligible.

(1 mark)

- (b) The forces acting on the ladder are in equilibrium.

Draw an arrow on **Figure 1** to show the direction of the resultant force from the ground acting on the ladder. Label your arrow **G**.

(2 marks)

- (c) The reaction at the ground acts at an angle of 60° to the horizontal. The ladder weighs 140 N.

Calculate the magnitude of the resultant force **G** from the ground acting on the ladder.

.....

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(2 marks)

- (d) If the wall was removed, only the weight **W** and reaction at the ground **G** would act, and the ladder would begin to topple.

State the direction of the resultant force **R** and calculate its magnitude.

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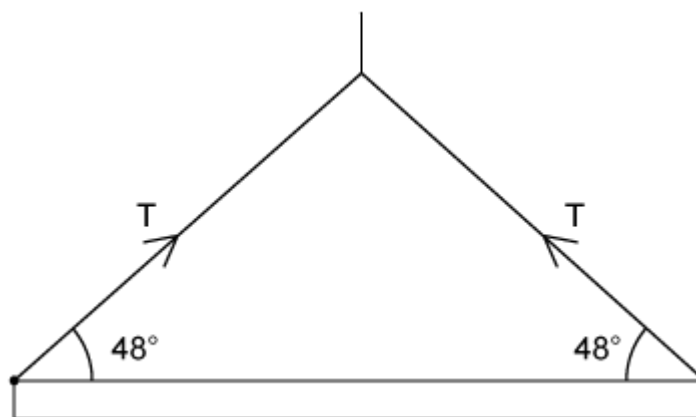
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(3 marks)

4 (a) **Figure 1** shows a uniform steel girder being held horizontally by a crane.

Two cables are attached to the ends of the girder and the tension in each of these cables is T .

Figure 1



On **Figure 1** draw an arrow to show the line of action of the weight of the girder.

(1 mark)

(b) The tension, T , in each cable is 650 N.

Calculate the horizontal component of the tension in each cable.

(2 marks)

(c) Calculate the vertical component of the tension in each cable.

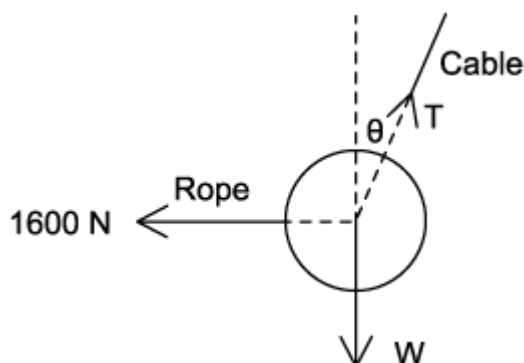
(2 marks)

(d) Calculate the weight of the girder.

(2 marks)

5 (a) **Figure 1** shows a 500 kg iron ball being used on a demolition site.

Figure 1



The ball is suspended from a cable which makes an angle θ with the vertical, and is pulled into the position shown by a rope that is kept horizontal. The tension in the rope is 1600 N.

Determine the magnitude of the vertical component of the tension in the cable and write an expression relating the angle θ with the vertical and T .

(3 marks)

(b) Determine the magnitude of the horizontal component of the tension in the cable and write an expression relating the angle θ with the vertical and T .

(3 marks)

(c) Determine the angle θ the cable makes with the vertical.

(3 marks)

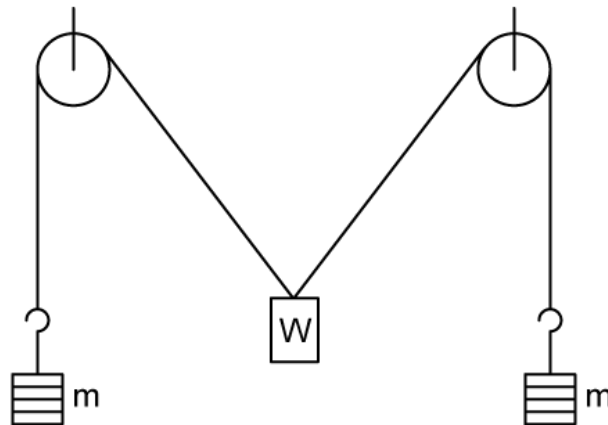
(d) Determine the magnitude of the tension in the cable.

(2 marks)

Hard Questions

- 1 (a)** On a construction site, a load, of weight, W , is supported by two strings kept in tension by equal masses, m , hung from their free ends, with each string passing over a frictionless pulley. The arrangement is symmetrical and is shown in **Figure 1**.

Figure 1



Draw a vector diagram for the three forces acting through the point at which the strings are attached to the load.

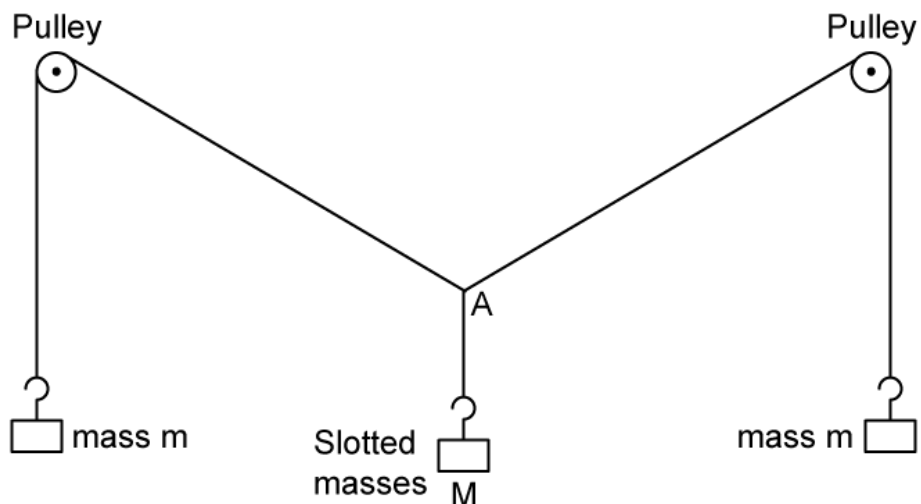
Label:

- The weight of the load, W
- The tensions in each string, in terms of m and g , the gravitational field strength
- The angle θ between each string and the vertical

(2 marks)

- (b)** The mass of the central load is M , as shown in **Figure 2**. The weight of the central load is still W .

Figure 2



By resolving the forces vertically about point A, determine an expression for:

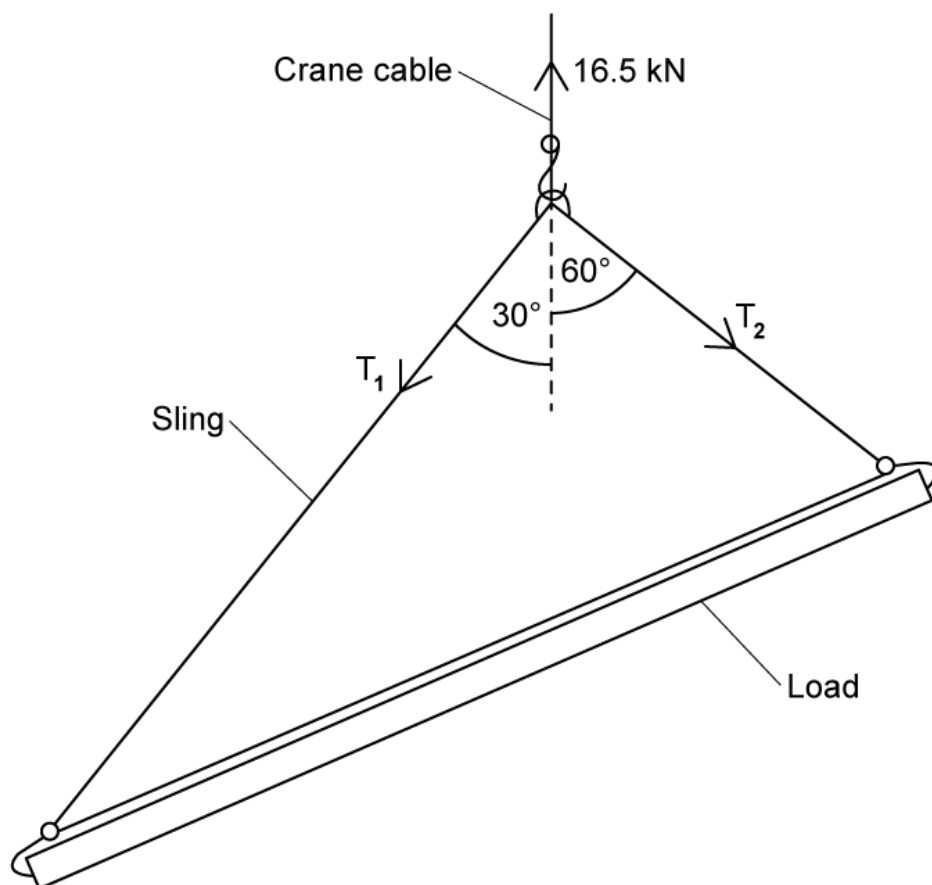
(i) m in terms of W , g and θ .

(ii) M in terms of m and θ .

(3 marks)

(c) **Figure 3** shows a crane hook in equilibrium under the action of a vertical force of 16.5 kN in the crane cable and tension forces T_1 and T_2 in the sling.

Figure 3

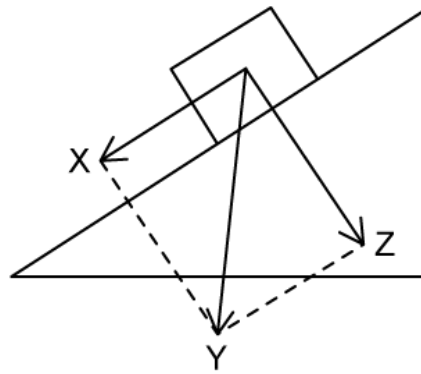


By drawing a scale diagram, and stating an appropriate scale, determine the tension forces T_1 and T_2 acting in the sling.

(4 marks)

(d) A crate rests on the inclined load as shown in **Figure 4**.

Figure 4



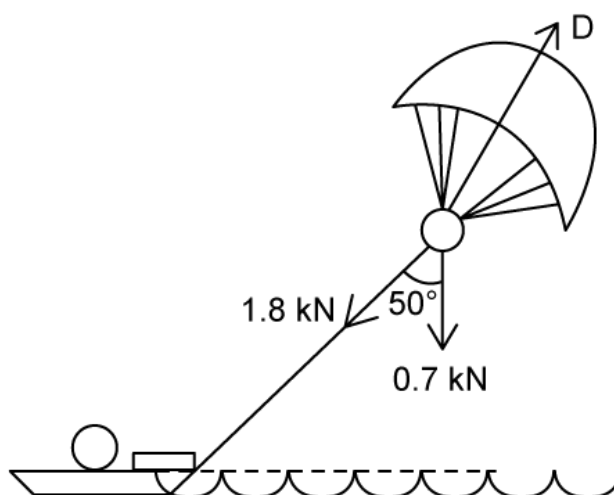
Explain the effects on vectors **X**, **Y** and **Z** if the incline becomes steeper.

(3 marks)

- 2 (a)** In the leisure activity parascending, a person attached to a parachute is towed over the sea by a tow rope attached to a motorboat.

Figure 1 shows the directions of the forces acting on a person of weight 0.7 kN when being towed horizontally at a constant speed. The 1.8 kN force is the tension in the tow rope and the force labelled **D** is the drag force.

Figure 1

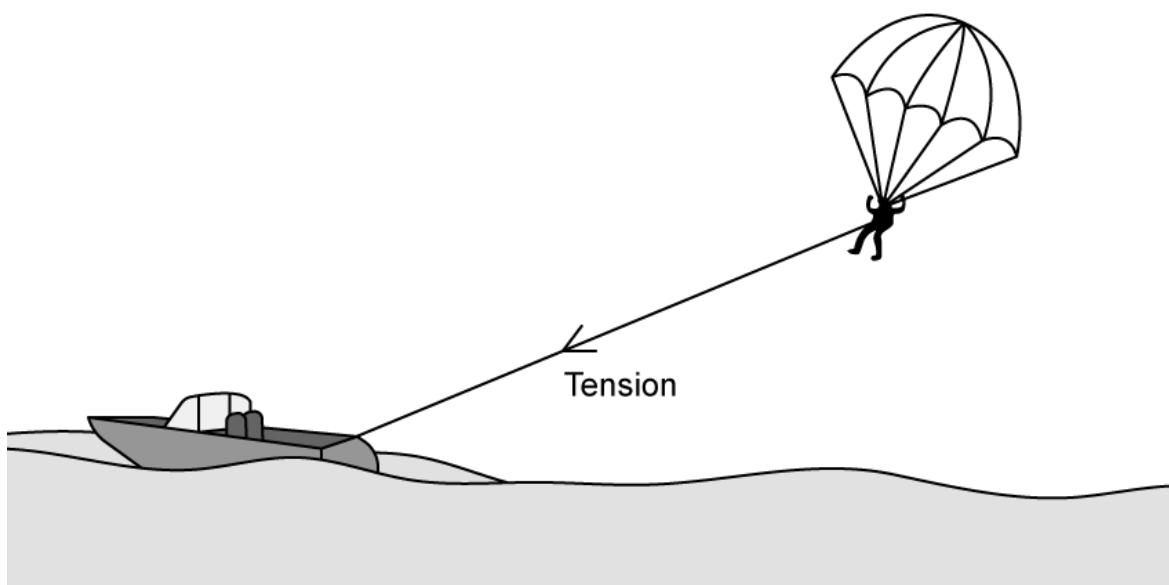


Draw a vector triangle relating the forces acting on the person. Hence, or otherwise, determine the magnitude of the drag force.

(4 marks)

- (b)** **Figure 2** shows a different parasailer being towed at a constant velocity.

Figure 2



The rope towing the parasailer makes an angle of 27° with the horizontal and has a tension of 2.2 kN. The drag force of 2.6 kN acts at an angle of 41° to the horizontal.

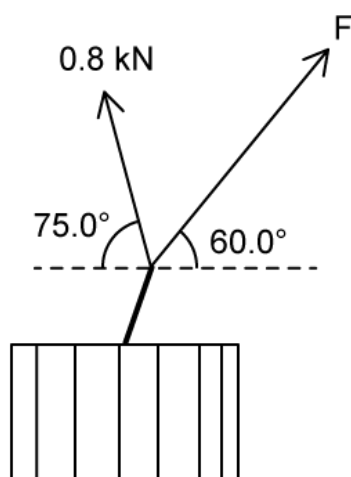
By drawing a scale diagram, and stating an appropriate scale, determine the weight of the parasailer.

(4 marks)

- (c) The rope breaks loose from the motorboat, but luckily two of the people on the boat grab it as it reaches an angle of 75° to the horizontal.

The two people pull the rope with different forces in different directions, as shown in the **Figure 3**, which shows a bird's eye view of the parasailer.

Figure 3



The first person pulls the rope with a force of 0.8 kN at an angle of 75.0° to the horizontal. The second person pulls at an angle of 60.0° to the horizontal.

Calculate the magnitude of the force F that the second person must pull the rope with in order to match the tension of the rope in part (b).

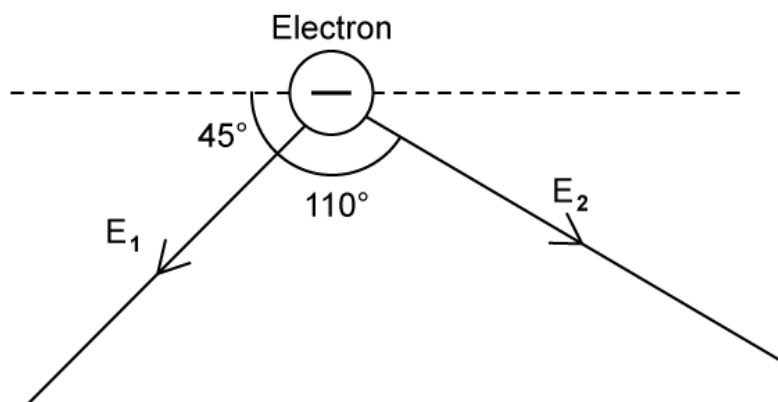
(3 marks)

- (d) Determine the magnitude and direction of the force that the boat would have to exert on the rope to make the resultant force equal to zero.

(1 mark)

- 3 (a)** An electron is under the influence of two electric fields, E_1 and E_2 , with values $3.0 \times 10^{-8} \text{ V m}^{-1}$ and $6.0 \times 10^{-8} \text{ V m}^{-1}$ respectively, as shown in **Figure 1**.

Figure 1



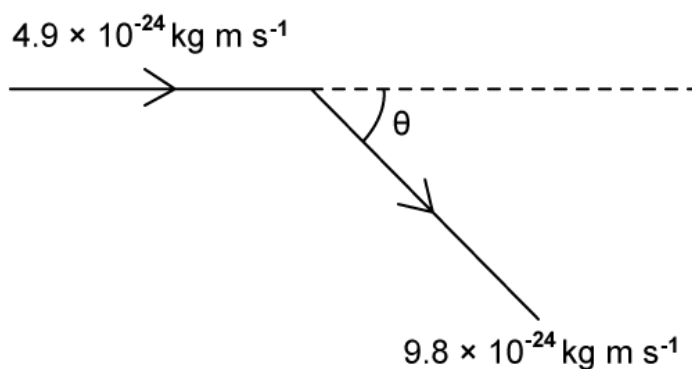
Determine the magnitude of the resultant field strength E acting on the electron and the angle it makes with the positive horizontal axis.

(3 marks)

- (b)** The electron has an initial momentum of $4.9 \times 10^{-24} \text{ kg m s}^{-1}$ in the horizontal direction.

Upon impact, its momentum is $9.8 \times 10^{-24} \text{ kg m s}^{-1}$ at an angle of θ to the horizontal, as shown in **Figure 2**.

Figure 2



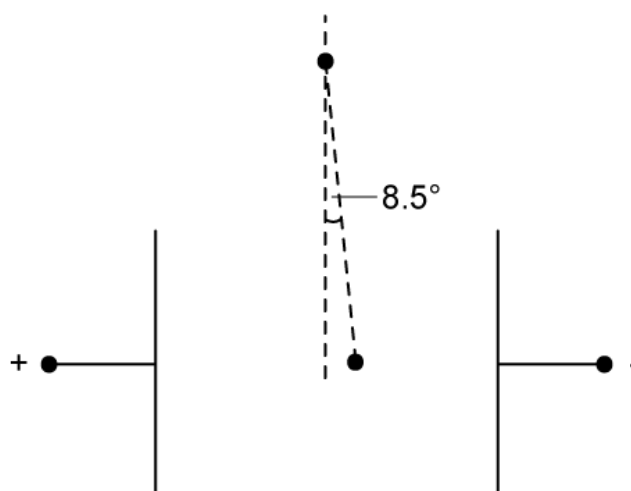
Determine the magnitude and direction of the change in momentum relative to the positive horizontal axis.

(2 marks)

- (c) A small, charged sphere, suspended from a thread of insulating material, was placed between two vertical parallel plates.

When a potential difference was applied to the plates, the sphere moved until the thread made an angle of 8.5° to the vertical, as shown in **Figure 3**.

Figure 3

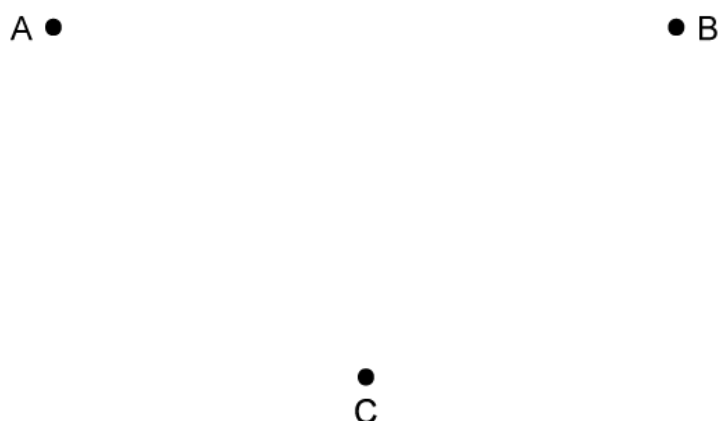


Show that the electrostatic force F on the sphere is given by $F = mg \tan 8.5^\circ$, where m is the mass of the sphere.

(3 marks)

(d) Point charges **A** and **B** are placed a distance apart, as shown in **Figure 4**.

Figure 4



The electric field is directed towards the positive charge and away from the negative charge.

A holds a positive charge and is half the charge of **B**. **B** holds a negative charge. Point **C** is equidistant from the two charges

(i) Draw two arrows on **Figure 4** at **C** to represent the directions and relative magnitudes of the components of the electric field at **C** due to each of the charges.

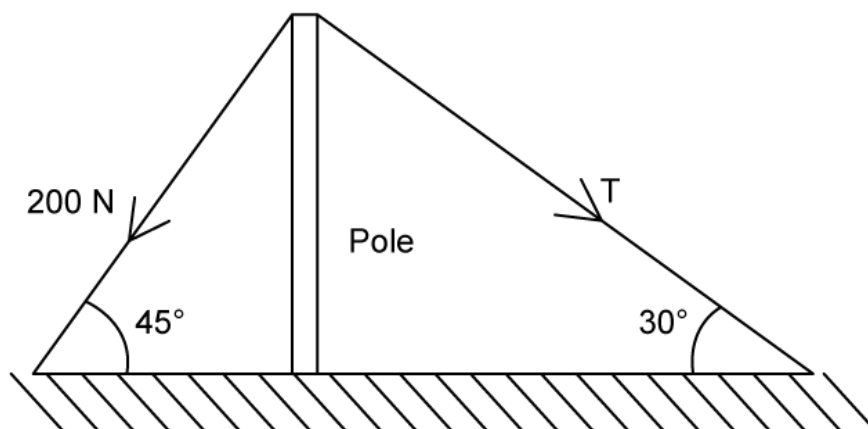
(ii) Hence draw an arrow, labelled **E**, on **Figure 4** at **C** to represent the direction of the resultant electric field at **C**.

(3 marks)

4 (a) **Figure 1** shows a pole being kept vertical to the ground by two taut light ropes.

The angles between the ropes and the ground are shown in **Figure 1**. The tension in the left rope is 200 N, and the tension in the right rope is T .

Figure 1



Use a scale diagram to determine the weight of the pole, W and the tension T .

(4 marks)

(b) A canoeist can paddle at a speed of 3.8 m s^{-1} in still water.

She encounters a current which opposes her motion. The current has a velocity of 1.5 m s^{-1} at 30° to her original direction of travel as shown in **Figure 2**.

Figure 2

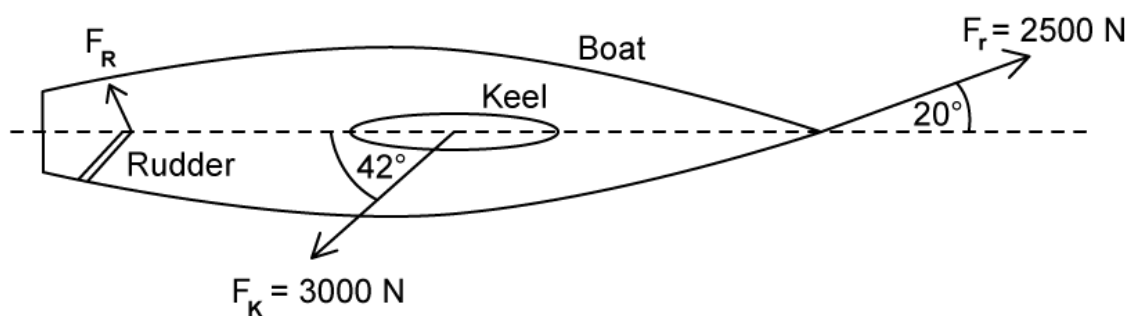


By drawing a scale diagram, and stating an appropriate scale, determine the magnitude of the canoeist's resultant velocity.

(3 marks)

- (c) The boat shown in **Figure 3** is being towed at constant velocity. The tension force (F_T) in the towing rope is 2500 N. The forces resisting the motion can be assumed to be the force of the water on the keel (F_K) and the force of the water on the rudder (F_R).

Figure 3



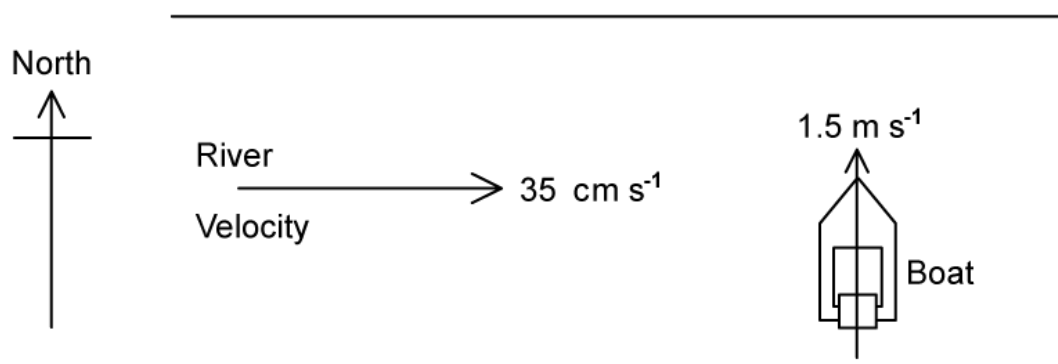
By calculation or by scale drawing, find the size of the force, F_R , needed to keep the boat in equilibrium.

(4 marks)

(d) **Figure 4** shows a river which flows from West to East at a constant velocity of 35 cm s^{-1} .

The boat leaves the south bank heading due North at 1.50 m s^{-1} .

Figure 4



Using a scale diagram, determine the resultant velocity of the boat, stating both direction and magnitude.

(4 marks)

- 5 (a)** A plane flying across the Lake District sets off from base camp to lake Windermere, 28 km away, in a direction of 20.0° north of east.

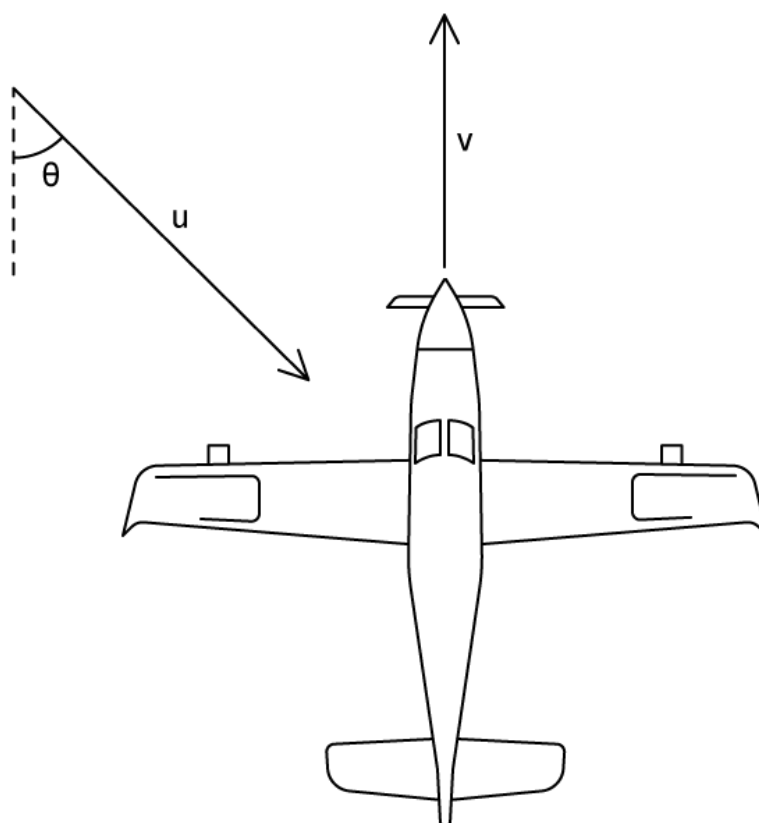
After dropping off supplies it flies to lake Coniston, which is 19 km at 30.0° west of north from lake Windermere.

Determine the distance from lake Coniston to base camp.

(3 marks)

- (b)** **Figure 1** shows the plane flying due north through still air with a speed v .

Figure 1



The plane enters a region where the wind is blowing with a speed u from a direction which makes an angle of θ with due south.

Write an expression for the time taken for the plane to fly a distance D due north of its current position in this windy region.

(2 marks)

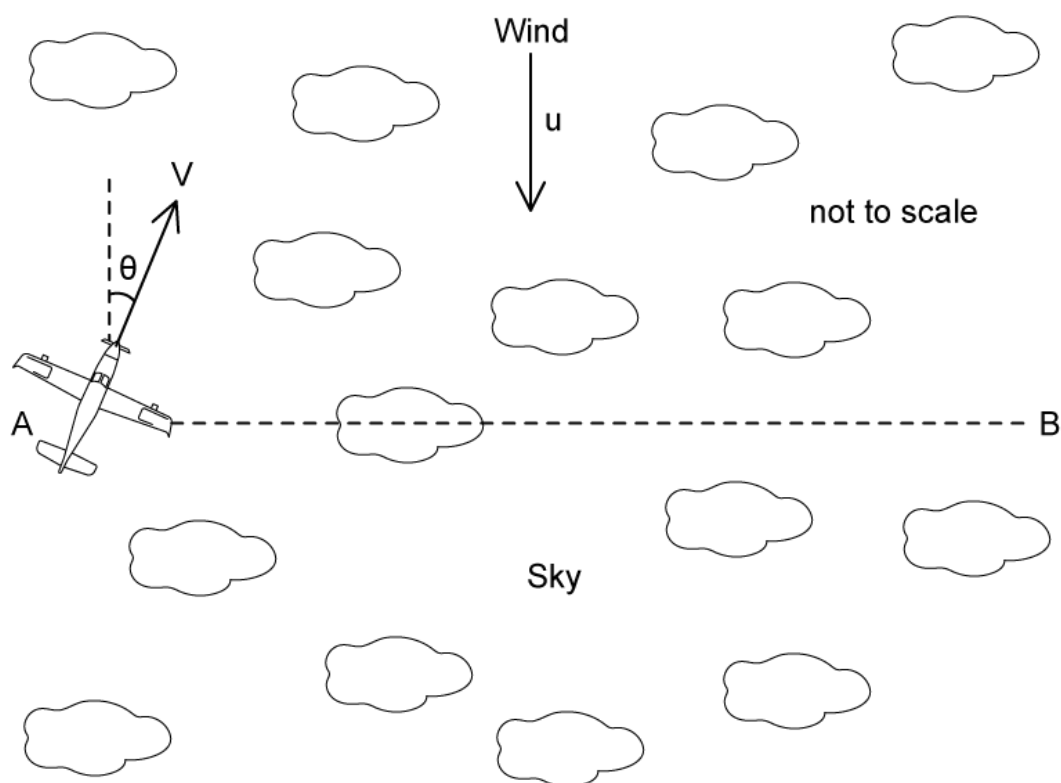
- (c)** When there is no wind, the aircraft travels 180 km every 30 minutes. When there is a wind blowing 53° south, the aircraft takes an extra 4 minutes to travel the same distance.

Assuming the orientation of the plane does not change, calculate the speed of the wind in km h^{-1} .

(3 marks)

- (d)** The pilot wishes to cross the sky in a straight line between two points labelled **A** and **B** as shown in **Figure 2**.

Figure 2



The wind is now blowing due south with the same speed as in part (c). The plane continues to travel at the same speed in the wind as in part (c).

To cross from **A** to **B** the pilot has to turn the plane at an angle θ to the direction of the wind, as shown in **Figure 2**.

Determine the size of the angle θ using a scale drawing.

(3 marks)